

Lecture Eight

1 Applications

1.1 Related Rates

Here we will study the theme of the of derivatives as rates of change in problems in which the variable changes with respect to time. The essential feature of these problems is that two or more variables which are related in a known way are themselves changing in time.

Here are two examples illustrating this type of problem.

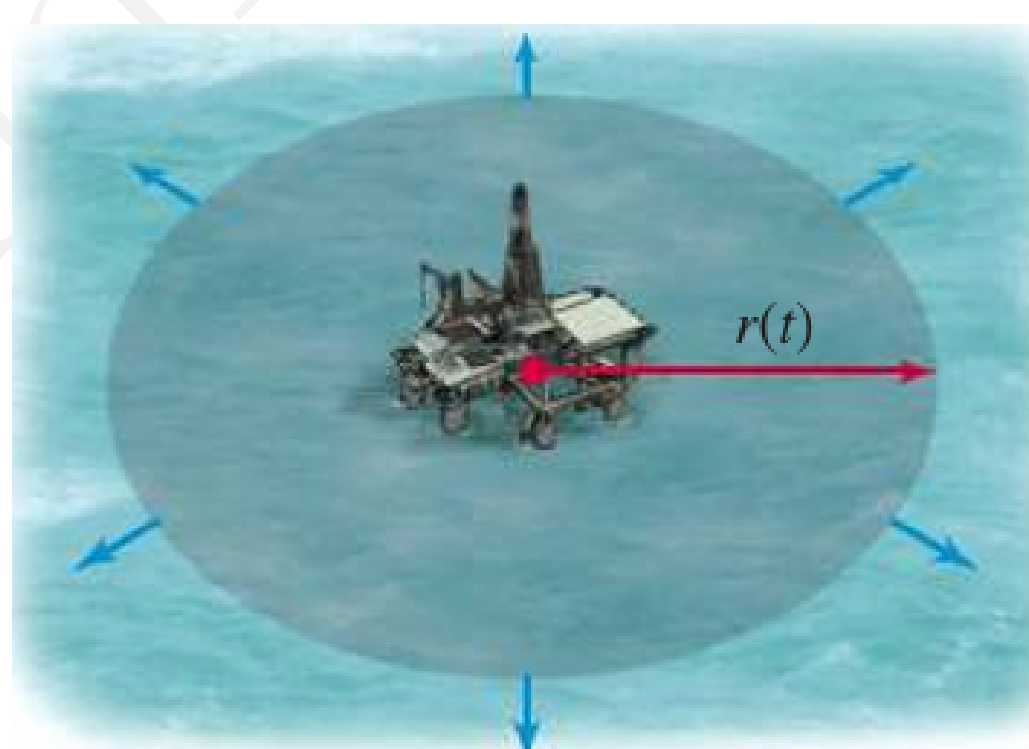
1. An oil rig springs a leak and the oil spreads in a (roughly) circular patch around the rig. If the radius of the oil patch increases at a known rate, how fast is the area of the patch changing?
2. Two airlines approach an airport with known speeds, one flying west and one flying north. How fast is the distance between the airlines changing?

In the first problem, the two related variables are the radius and the area of the oil patch, both are changing in time. But the second problem has three related variables, the positions of the two airlines and the distance between them.

The goal in both problems is to determine the rate of change of one of the variables at a specific moment of time, hence the name **related rates**.

Example 1.1.1 Spreading Oil: *An oil rig springs a leak in calm seas, and the oil spreads in a circular patch around the rig. If the radius of the oil patch increases at a rate of 30 m/hr, how fast is the area of the patch increasing when the patch has a radius of 100 meters.*

Sol



The key relationship between the radius and the area is $A = \pi r^2$. The general expression relating the radius and area at any time t is

$$A(t) = \pi r^2(t) \quad (1.1.1)$$

. The goal is to find the rate of change of the area of the circle, which $A'(t)$ given that $r'(t) = 30$ m/hr.

We differentiate both sides of the equation 1.1.1 w.r.t time t .

$$A'(t) = \frac{d}{dt}[\pi r^2(t)] = 2\pi r(t).r'(t)$$

Now by substituting the given values $r(t) = 100$ m and $r'(t) = 30$ m/hr we have (including units):

$$A'(t) = 2\pi r(t).r'(t) = 2\pi(100 \text{ m})(30 \text{ m/hr}) = 6000\pi \text{ m}^2/\text{hr}$$

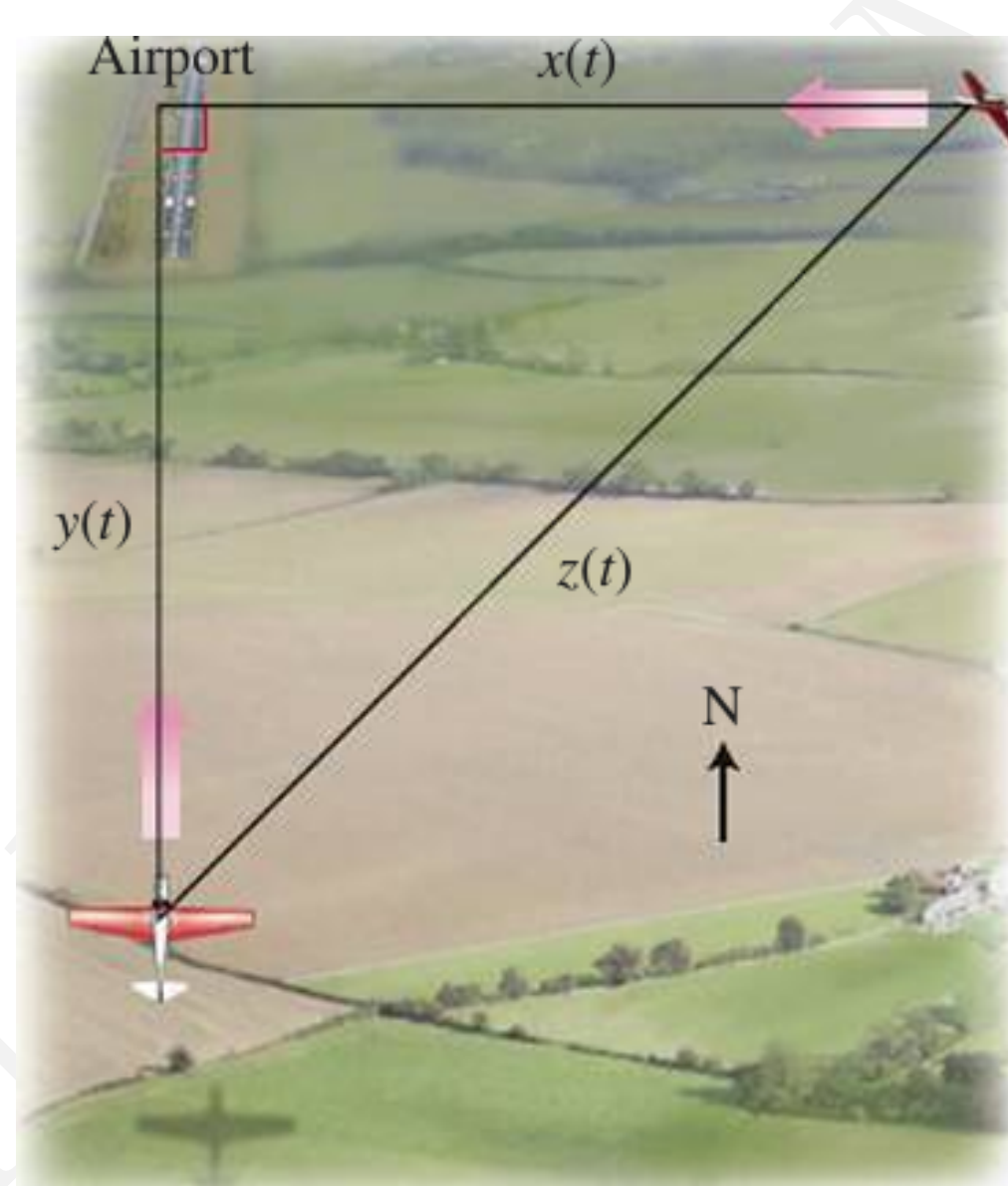
1.1.1 Procedure Steps for Related Rates Problems

1. Read the problem carefully, making a sketch to organize the given information. Identify the rates that are given and the rate that is to be determined.
2. Write one or more equations that express the basic relationships among the variables.
3. Introduce rates of change by differentiating the appropriate equation(s) with respect to time (t).
4. Substitute known values and solve for the desired quantity.
5. Check that the units are consistent and the answer is reasonable.

Example 1.1.2 Converging Air-planes:

Two small planes approach an airport, one flying due west at 120 mi/hr and the other flying due north at 150 mi/hr. Assuming they fly at the same constant elevation, how fast is the distance between the planes changing when the west bound plane is 180 miles from the airport and the north bound plane is 225 miles from the airport?

Sol



Let $x(t)$ and $y(t)$ denote the distance from the airport to the west bound and north bound planes respectively. The paths of the two planes form the legs of a right triangle and the distance between them denoted $z(t)$. By Pythagorean theorem $z^2 = x^2 + y^2$. Now

$$\frac{d}{dt}[z^2] = \frac{d}{dt}[x^2 + y^2] \Rightarrow 2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt} \Rightarrow \boxed{\frac{dz}{dt} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{z}}$$

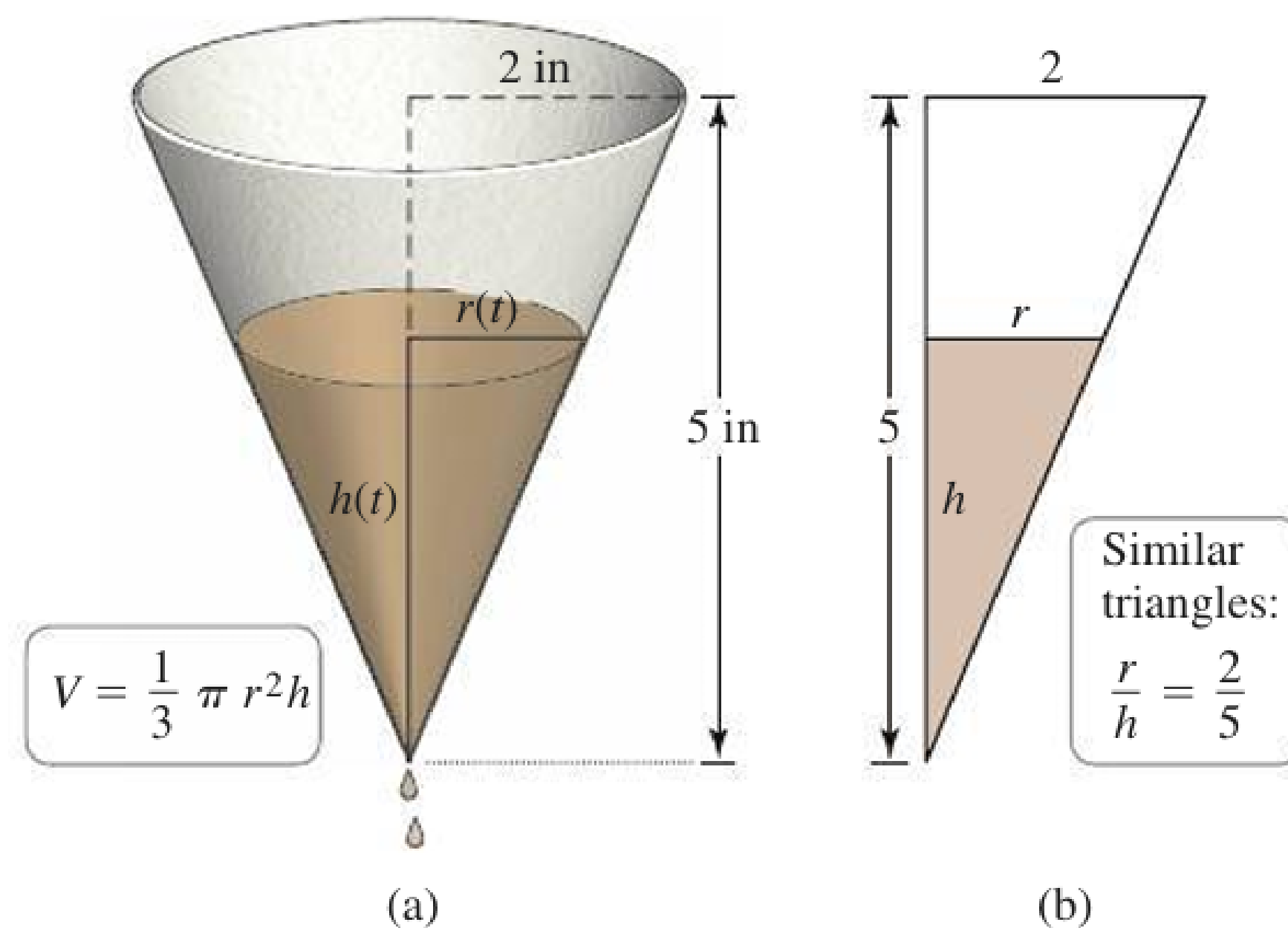
Given $\frac{dx}{dt} = -120$ mi/hr, $\frac{dy}{dt} = -150$ mi/hr and $x = 180$ mi, $y = 225$ mi/hr. The distance between the planes is $z = \sqrt{x^2 + y^2} = \sqrt{(180)^2 + (225)^2} = 288$ mi. Now:

$$\frac{dz}{dt} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{z} = \frac{(180)(-120) + (225)(-150)}{288} = -192 \text{ mi/hr.}$$

Example 1.1.3 Morning Coffee:

Coffee is draining out of a conical filter at a rate of $2.25 \text{ in}^3/\text{min}$. If the cone is 5 in tall and has a radius of 2 in, how fast is the coffee level dropping when the coffee level is 3 in deep?

Sol



We want to find $\frac{dh}{dt}$ at $h = 3$ given that $\frac{dv}{dt} = -2.25 \text{ in}^3/\text{min}$. The volume formula for a cone is

$$v = \frac{1}{3}\pi r^2 h$$

By using similar triangles the ratio of the radius of the coffee in the cone to the height of the coffee is $\frac{2}{5}$ at all time.

$$\Rightarrow \frac{r}{h} = \frac{2}{5} \text{ or } r = \frac{2}{5}h$$

Substituting $r = \frac{2}{5}h$ in the volume formula gives V in terms of h

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{2}{5}h\right)^2 h = \frac{4\pi}{75}h^3 \quad (1.1.2)$$

Now differentiating both sides of 1.1.2 w.r.t t , we get

$$V'(t) = \frac{4\pi}{25}h^2(t) \cdot \frac{dh}{dt}$$

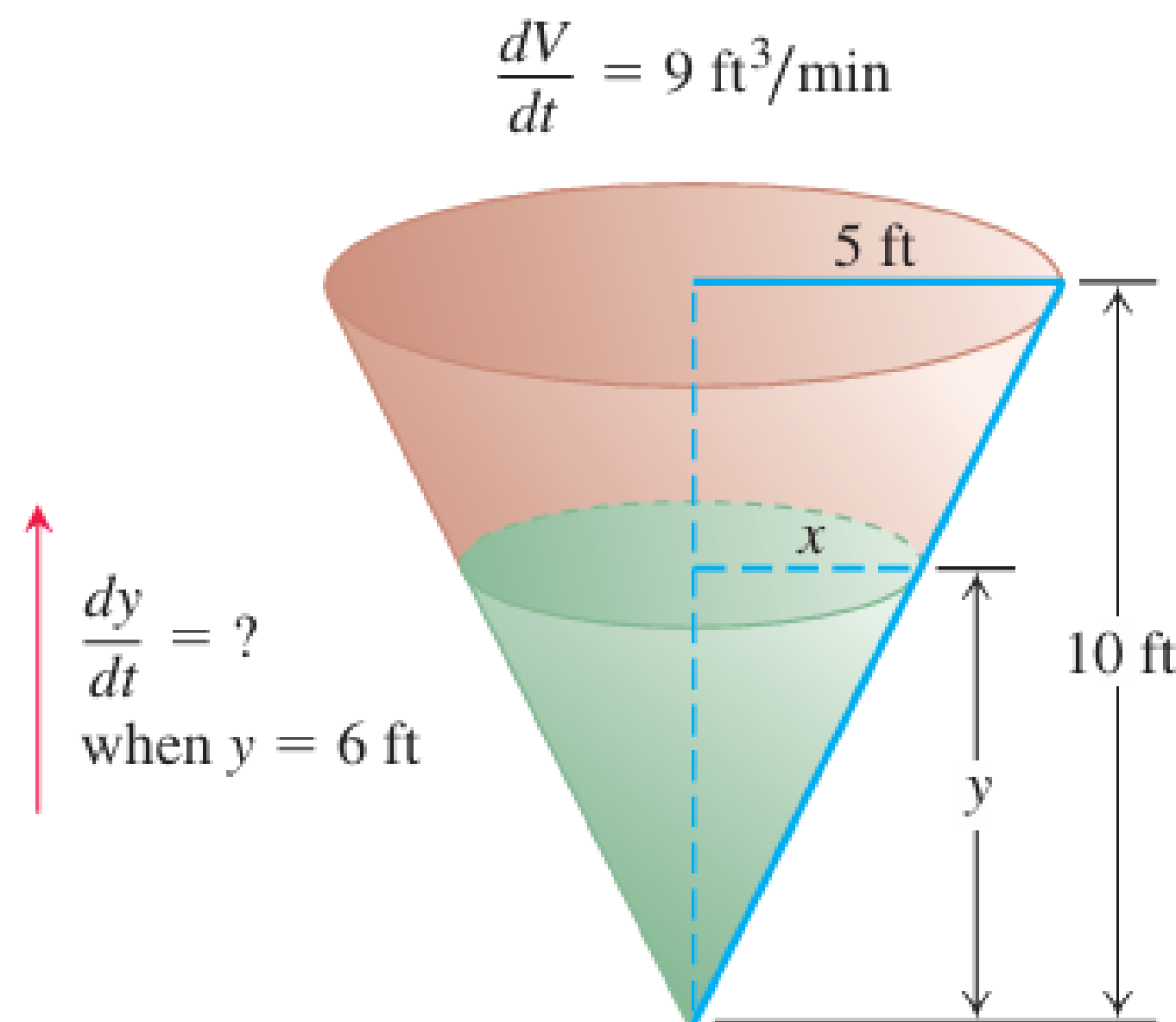
Now, we find $\frac{dh}{dt}$ at the instant when $h = 3$, given that $V'(t) = -2.25 \text{ in}^3/\text{min}$. Solving for $\frac{dh}{dt}$ and substituting these values, we have

$$\frac{dh}{dt} = \frac{25}{4\pi h^2(t)} V'(t) = \frac{25(-2.25)}{4\pi(3)^2} = -0.497 \text{ in/min}$$

It means, at the instant that the coffee is 3 in deep, the height decreases at a rate of 0.497 in/min (almost half an inch per min).

Example 1.1.4 Water runs into a conical tank at the rate of $9 \text{ ft}^3/\text{min}$. The tank stands point down and has a height of (10 ft) and a base radius of (5 ft). How fast is the water level rising when the water is (6 ft) deep?

Sol



The variables in the problem are:

$V \equiv$ Volume (ft^3) of the water in the tank at time t
 $x \equiv$ radius (ft) of the surface of the water at time t .
 $y \equiv$ depth (ft) of the water in the tank.

V, x and y are differentiable functions of t . We asked for $\frac{dy}{dt}$ when $y = 6 \text{ ft}$ and $\frac{dV}{dt} = 9 \text{ ft}^3/\text{min}$.

The water forms a cone with volume $V = \frac{1}{3}\pi x^2 y$.

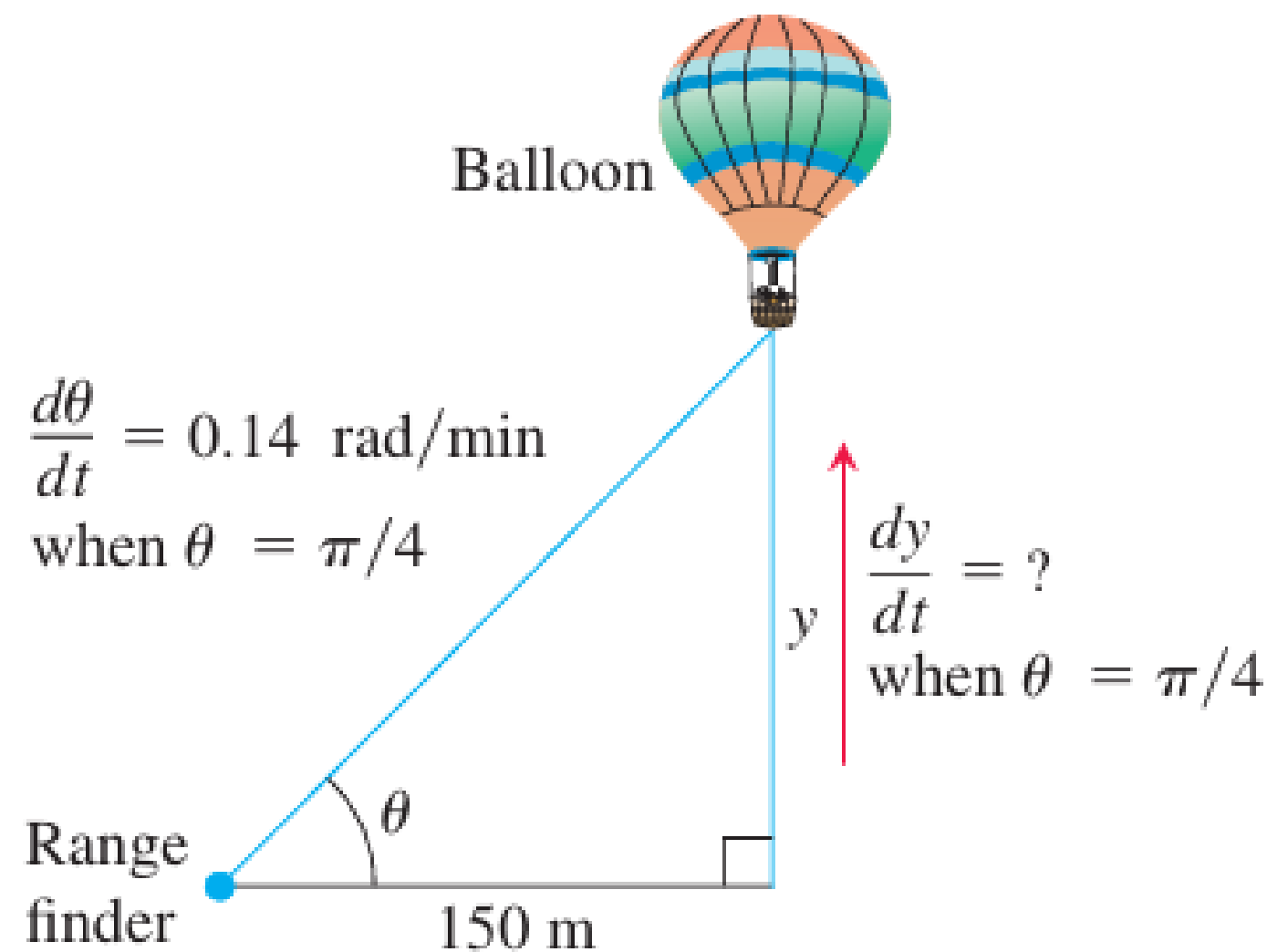
Because no information given about x and $\frac{dx}{dt}$ at the time in question, we need to eliminate x . The similar triangles gives us a way to express x in terms of y as $\frac{x}{y} = \frac{5}{10} \Rightarrow x = \frac{y}{2}$.

Therefore, we find $V = \frac{1}{3}\pi \left(\frac{y}{2}\right)^2 y = \frac{\pi}{12}y^3$. Now $\frac{dV}{dt} = \frac{\pi}{4}y^2 \frac{dy}{dt}$. Finally, we use $y = 6$ and $\frac{dV}{dt} = 9$.

$$9 = \frac{\pi}{4}(6)^2 \frac{dy}{dt} \Rightarrow \frac{dy}{dt} = \frac{1}{\pi} \approx 0.32$$

Example 1.1.5 A hot air balloon rising straight up a level field is tracked by a range finder 150 m from the left off point. At the moment the range finder's elevation angle is $\frac{\pi}{4}$, the angle increasing at the rate of 0.14 rad/min. How fast is the balloon rising at the moment?

Sol



The variables in this problem are:

$\theta \equiv$ The angle in radians that the range finder makes with the ground.
 $y \equiv$ The height in meters of the balloon above the ground.

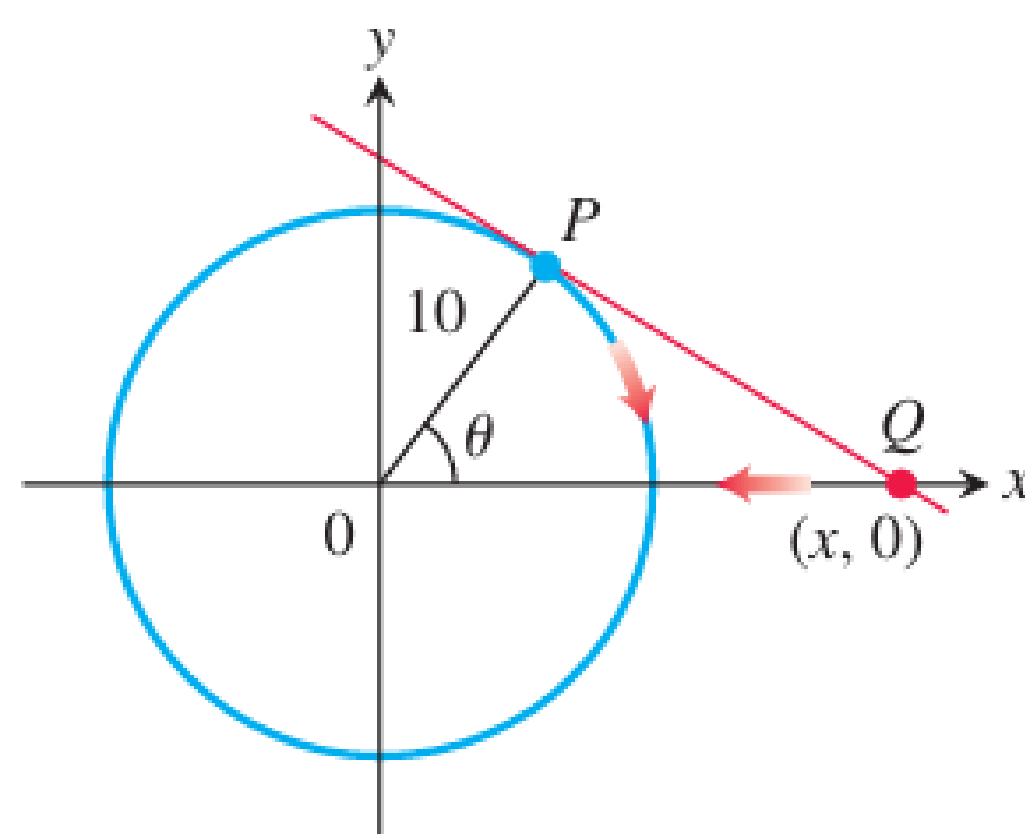
Given $\frac{d\theta}{dt} = 0.14$ rad/min when $\theta = \frac{\pi}{4}$. We want $\frac{dy}{dt}$ when $\theta = \frac{\pi}{4}$, and we know from the right triangle $\tan \theta = \frac{y}{150}$ or $y = 150 \tan \theta$, this implies that:

$$\frac{dy}{dt} = 150 \sec^2 \theta \frac{d\theta}{dt} \Rightarrow \frac{dy}{dt} = 150 \left[\sec^2 \frac{\pi}{4} \right] (0.14) = 42 \text{ m/min.}$$

At the moment in question, the balloon is rising at the rate of 42 m/min.

Example 1.1.6 A particle p moves clockwise at a constant rate along a circle of radius 10 m centred at the origin. The particle's initial position is $(0, 10)$ on the y -axis, and its final destination is the point $(10, 0)$ on the x -axis. Once the particle is in motion, the tangent line at p intersects the x -axis at a point Q (which moves over time). If the particle takes 30 sec to travel from start to finish, how fast is the point Q moving along the x -axis when it is 20 m from the centre of the circle?

Sol



We picture the situation in the coordinate plane with circle centred at the origin

$$t \equiv \text{time}$$

$$\theta \equiv \text{the angle from the } x - \text{axis to the radial line joining the origin to } p$$

Since the particle travels from start to finish in 30 sec, it travels along the circle at a constant rate of $\frac{\pi}{2}$ rad in $\frac{1}{2}$ min or π rad/min $\Rightarrow \frac{d\theta}{dt} = -\pi$ where t measured in min and the negative sign because θ is decreasing over time.

Setting $x(t)$ to be the distance at time t from the point Q to the origin. We want to find $\frac{dx}{dt}$ when $x = 20$ m, $\frac{d\theta}{dt} = -\pi$ rad/min. To relate the variables x and θ , we see $x \cos \theta = 10$ or $x = 10 \sec \theta$.

$$\Rightarrow \frac{dx}{dt} = 10 \sec \theta \tan \theta \frac{d\theta}{dt} = -10\pi \sec \theta \tan \theta$$

When $x = 20 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \sec \theta = 2$ and $\tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{3}$. Now

$$\frac{dx}{dt} = (-10\pi) (2\sqrt{3}) = -20\sqrt{3} \pi.$$

At the moment in question, the point Q is moving toward the origin at the speed of $20\sqrt{3}\pi \approx 109$ m/min.